EPFL ETP

River Flow Meshing Report

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1 Meshing strategies

Create the *Named Selections* for the boundary

- Rhine_in for the inlet of the Rhine
- Neckar_in for the inlet of the Neckar
- Rhine_out for the outlet of the Rhine
- Wall_1, Wall_2, Wall_3 for the three riverbanks

With the mesh selected in the project tree, set *Physics Preference* to CFD and *Element Size* to 15m under **Details** → **Defaults**. Add an *Inflation* and a *Method* property to the mesh, as well as an *Edge Sizing* for all 6 edges (3 walls, 2 inlets, 1 outlet). Under **Details** → **Definition**, after having clicked on inflation in the project tree, set *Boundary Scoping Method* to Named Selections and *Boundary* to Wall_1 & Wall_2 & Wall_3 by holding down ctrl. We also increase *Maximum Layers* to 4 to get a finer mesh near the borders without altering the inner regions of the mesh too much. For each edge sizing, under **Details** → **Definition**, set *Type* to Factor of Global Size.

The meshing parameters to act upon are the *Method*, the inflation *Growth Rate*, and the edge sizing *Element Size Factor* for each boundary line. To determine a satisfactory combination of parameters, we use the Taguchi method. In order to keep the number of required experiments reasonably low, we use one element size factor per boundary type (walls vs. in/outlets). We differentiate them due to the significant difference in scale. We use the Taguchi L9 orthogonal array, which is suitable for four 3-level factors.

	1	2	3	4
1	1	1	1	1
2	1	2	2	2
3	1	3	3	3
4	2	1	2	3
5	2	2	3	1
6	2	3	1	2
7	3	1	3	2
8	3	2	1	3
9	3	3	2	1

Table 1: Taguchi L9

The values to be tested for each parameter are chosen to be realistic and reasonable while being different enough to have a noticeable impact on the resulting mesh, in order to have a meaningful evaluation and comparison. Ultimately, this somewhat comes down to trial-and-error and empirical judgment.

Parameter	Levels		
	1	2	3
method	Quadrilateral Dominant	Triangles	MultiZone Quad/Tri
inflation growth rate	1.2	1.4	1.6
walls element size factor	0.5	0.75	1.0
in/outlets element size factor	0.5	0.75	1.0

Table 2: Meshing parameters design levels

The desired mesh quality metric can be set under **Details** → **Quality** → *Mesh Metric* after having selected the mesh in the project tree. The minimum, maximum, average and standard deviation of the metric can be viewed there as well. To avoid complications with multi-objective optimization, we solely maximize the average value, while simply monitoring the min and std. More information about the mesh quality metric can be found in section 3. The algorithm to determine the optimal parameters combination is implemented in the file *taguchi.py*

Experiment	Mesh quality metric		
	mean	std	min
1	0.98564	3.2559e-2	0.61695
2	0.99368	2.6767e-2	0.57822
3	0.99283	3.2757e-2	0.45389
4	0.97398	4.0407e-2	0.31426
5	0.97323	4.0123e-2	0.21215
6	0.97283	3.9929e-2	0.51198
7	0.99356	2.4473e-2	0.64168
8	0.98953	2.0809e-2	0.8283
9	0.99132	2.3951e-2	0.72091

Table 3: Taguchi design experiments results

Meshing method	**Inflation growth**	**Wall size factor**	**Inlet/outlet size factor**
MultiZone Quad/Tri	1.6	1.00	0.75

Table 4: Optimized meshing parameters

Another parameter we could have acted upon is the *Bias Type* on all edge sizing properties. We tried using the — – - – — bias type on the inner wall between the Rhine and the Neckar, with the intention of increasing the mesh density near the junction, which is the most critical zone in the system. This however created distortions in the mesh, that arise because cells across the river from one another are forced to have different sizes. This effect also prevents us from introducing a bias on the other river banks, since the lines are not of the same length, and therefore cannot match. A solution to this problem would be to divide the walls into shorter

line, granting finer control on the biases and element sizes to try to achieve a more uniform meshing. Biases on the ports boundaries did not do any good either.

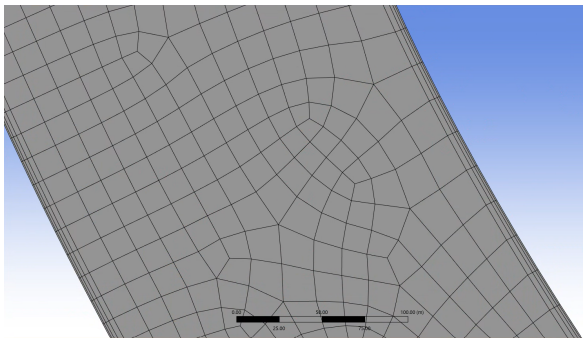


Figure 1: Mesh snapshot with biased inner wall (*Bias Factor* = 5)

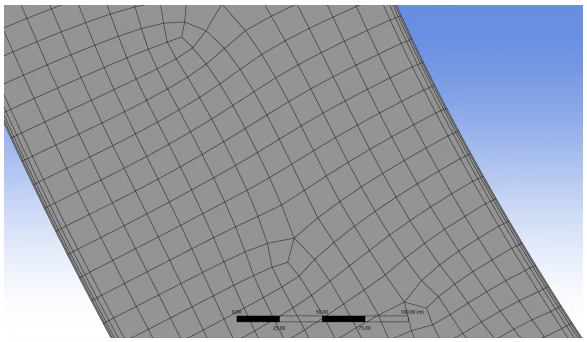


Figure 2: Mesh snapshot with no bias

2 Generated mesh

The optimization procedure settled for the MultiZone Quad/Tri method. It is the most flexible method, allowing the meshing algorithm to use both quadrilaterals and triangles. In our case, the mesh is composed solely of quadrilaterals (with most being close to squares), which coincides with our orthogonality requirements. The ports element size factor is less than that of walls, apriori to accommodate for the smaller spatial scale. The inflation growth rate is the highest among the possible options, resulting in a smoother transition between the tiny border cells and the larger inner cells.

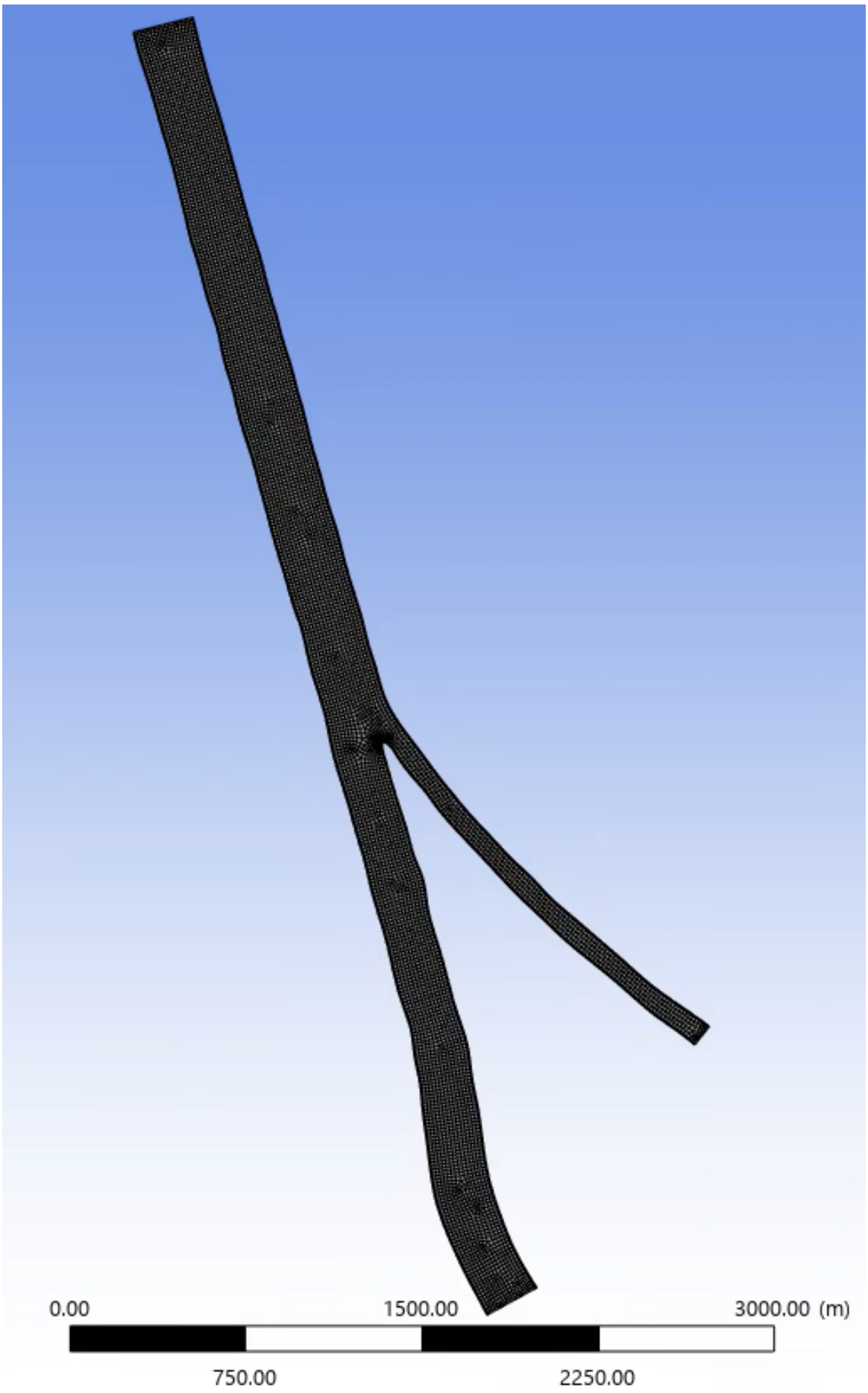


Figure 3: Visual representation of the generated mesh (global view)

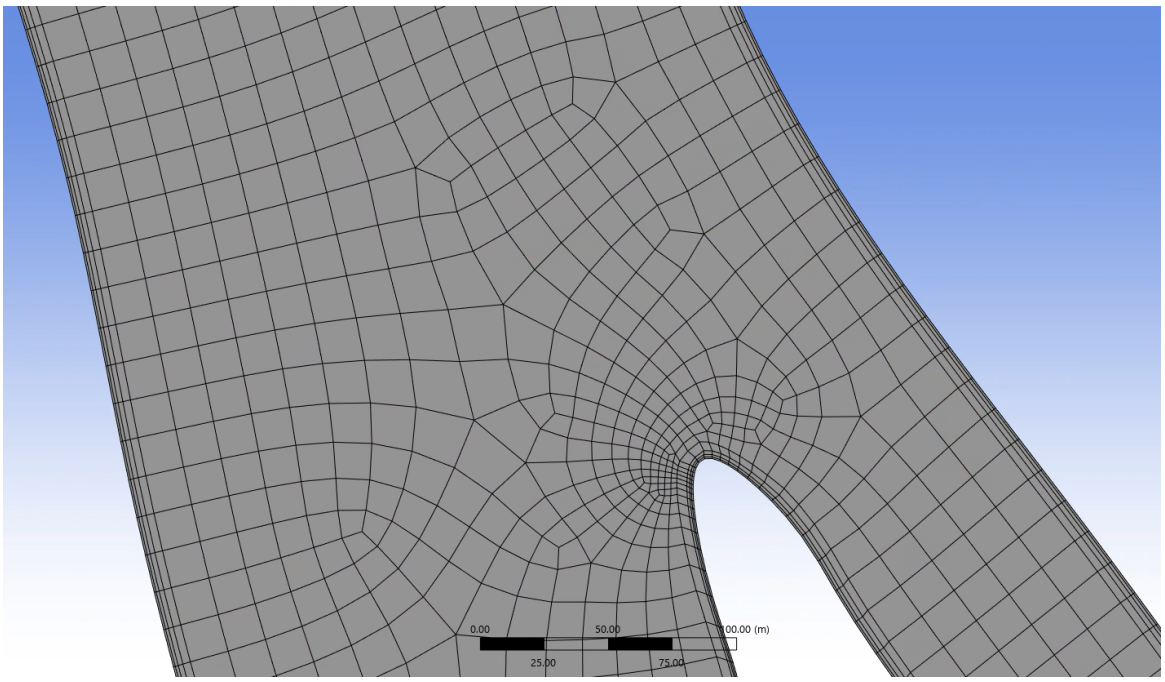


Figure 4: Visual representation of the generated mesh (junction)

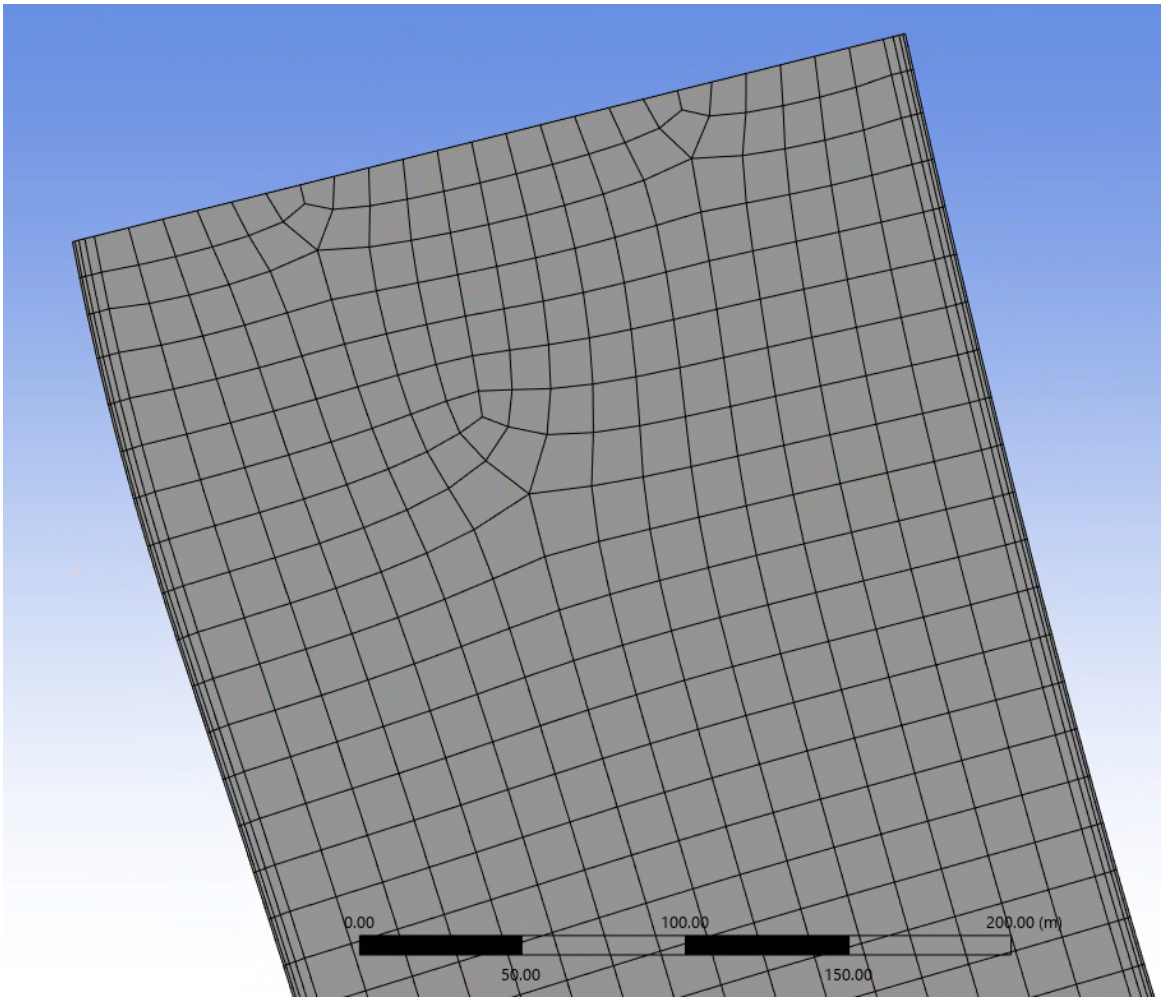


Figure 5: Visual representation of the generated mesh (outlet)

3 Mesh quality assessment

The mesh quality metric we chose is Orthogonal Quality, which ranges from 0 (worst) to 1 (best) for every cell. The official Ansys help page on mesh requirements and consideration cites it as one of the most important metric in Fluent. Orthogonality ensures better estimations of gradients and fluxes across cells, which is especially relevant for our confluent rivers flow modeling, where gradients near borders and the junction vary rapidly. The help page states that "the minimum orthogonal quality for all types of cells should be more than 0.01, with an average value that is significantly higher."

Min	**Max**	**Mean**	**Std**
0.7277	1.0000	0.99485	2.1782×10^{-2}

Table 5: Mesh quality assessment results through orthogonal quality

The average and minimum values are higher than any value encountered in Taguchi design experiments, meaning that single-objective optimization of the mean was an acceptable choice. Critically, we note that the minimum value is vastly satisfactory relative to the recommendations.